

Two-Limb Inductor with Saturation

Simscape model `twolimb_inductor_with_saturation`

Rev. 00 — 2026-09-19

Davide Bagnara

Rev.	Date	Description
00	2026-09-19	First issue

1 Scope

This note describes the Simscape component `twolimb_inductor_with_saturation`: a single-phase reactor with two windings on a common two-limb core, one winding in each conductor of a two-wire line. It is the two-winding counterpart of the `singlephase_inductor_with_saturation` block: same parameters, same saturation table, half of the inductance and of the resistance placed in each conductor. Typical uses are DC-link reactors split between the positive and negative rail, output reactors of thyristor rectifiers split in both poles, and single-phase AC reactors wound on two limbs.

The note gives the magnetic model (Sec. 2), the saturation characteristic (Sec. 3), the equivalence with the single-phase block (Sec. 4), the implementation details (Sec. 5), an equation-level verification (Sec. 6) and the limitations (Sec. 7). The source listing is in Appendix A.

2 Physical model

2.1 Magnetic circuit and winding connection

Fig. 1 shows the reactor: a two-limb core with an air gap in each limb and one coil of N turns on each limb. Coil 1 is connected between the terminals a+ and b+, coil 2 between a− and b−. Both winding currents are referred from the a side to the b side: i_1 flows from a+ to b+, i_2 from a− to b−. The load current enters at a+ and returns at a−, so in normal operation $i_1 = -i_2$.

The winding sense is chosen so that the load current magnetises the core: the magnetomotive force along the main loop is

$$\mathcal{F} = Ni_1 - Ni_2 = 2N i_m, \quad i_m = \frac{i_1 - i_2}{2}, \quad (1)$$

where i_m is the magnetising current. For the load current, $i_1 = -i_2 = i$, $i_m = i$; for a common-mode current, $i_1 = i_2$, $i_m = 0$ and the core carries no main flux.

Two flux paths are considered. The main path, with reluctance $\mathcal{R}_m$ (air gaps plus iron), links both coils:

$$\Phi_m = \frac{2N i_m}{\mathcal{R}_m}. \quad (2)$$

The leakage path of each coil, with reluctance $\mathcal{R}_\sigma$ (air only), links that coil alone. The flux linkages of the two coils are then

$$\lambda_1 = \frac{L_d}{2} i_1 + \frac{\psi}{2}, \quad \lambda_2 = \frac{L_d}{2} i_2 - \frac{\psi}{2}, \quad (3)$$

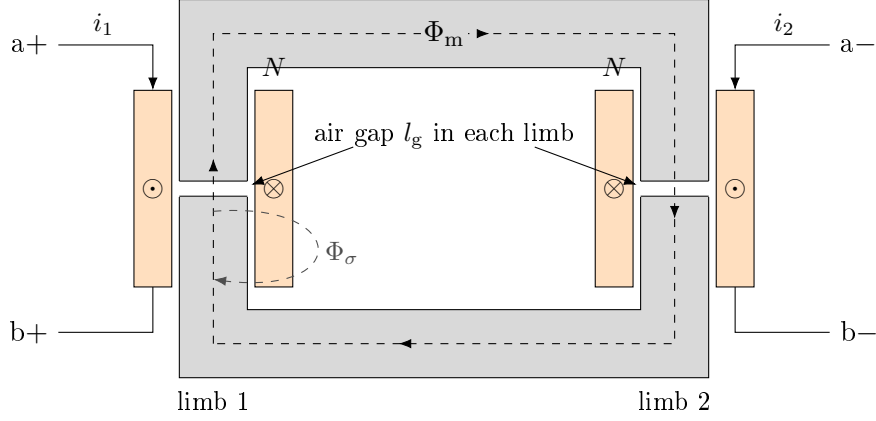


Figure 1: Two-limb reactor: one coil per limb, air gap in each limb. The current symbols ($\odot$ out of the page, $\otimes$ into the page) are drawn for the load current, $i_1 > 0$ and $i_2 < 0$: both coils drive the main flux Φ_m in the same direction around the core. Φ_σ is the leakage flux of coil 1, closing through the window and not linking coil 2.

with

$$\psi = 2N\Phi_m = \frac{4N^2}{\mathcal{R}_m} i_m = L i_m, \quad \frac{L_d}{2} = \frac{N^2}{\mathcal{R}_\sigma}. \quad (4)$$

ψ is the flux linkage of the main path for the whole loop (both coils in series), $L = 4N^2/\mathcal{R}_m$ is the main inductance seen by the load current and L_d the total leakage inductance seen by the load current. The sign of $\psi/2$ in λ_2 follows from the winding sense of coil 2 and from the reference direction of i_2 (a– to b–). L , L_d and the winding resistance R are the block parameters; they are *total* values, and half of each is assigned to each conductor.

2.2 Voltage equations

With $R/2$ per coil and the saturation model of Sec. 3, which replaces $\psi = L i_m$ by $d\psi = L k_L(i_m) di_m$, the terminal voltages u_1 (from a+ to b+) and u_2 (from a– to b–) are

$$u_1 = \frac{R}{2} i_1 + \frac{L_d}{2} \frac{di_1}{dt} + \frac{1}{2} \frac{d\psi}{dt}, \quad (5)$$

$$u_2 = \frac{R}{2} i_2 + \frac{L_d}{2} \frac{di_2}{dt} - \frac{1}{2} \frac{d\psi}{dt}, \quad (6)$$

$$\frac{d\psi}{dt} = L k_L(i_m) \frac{di_m}{dt} = L k_L(i_m) \frac{1}{2} \left(\frac{di_1}{dt} - \frac{di_2}{dt} \right). \quad (7)$$

These three equations are what the block implements. Fig. 2 shows the equivalent circuit: each conductor carries $R/2$, the uncoupled leakage $L_d/2$ and one half of the coupled (saturable) part.

2.3 Differential-mode and common-mode behaviour

With the mode currents and voltages

$$i_d = i_m = \frac{i_1 - i_2}{2}, \quad i_c = \frac{i_1 + i_2}{2}, \quad u_d = u_1 - u_2, \quad u_c = \frac{u_1 + u_2}{2}, \quad (8)$$

eqs. (5)–(7) separate into

$$u_d = R i_d + [L_d + L k_L(i_d)] \frac{di_d}{dt}, \quad (9)$$

$$u_c = \frac{R}{2} i_c + \frac{L_d}{2} \frac{di_c}{dt}. \quad (10)$$

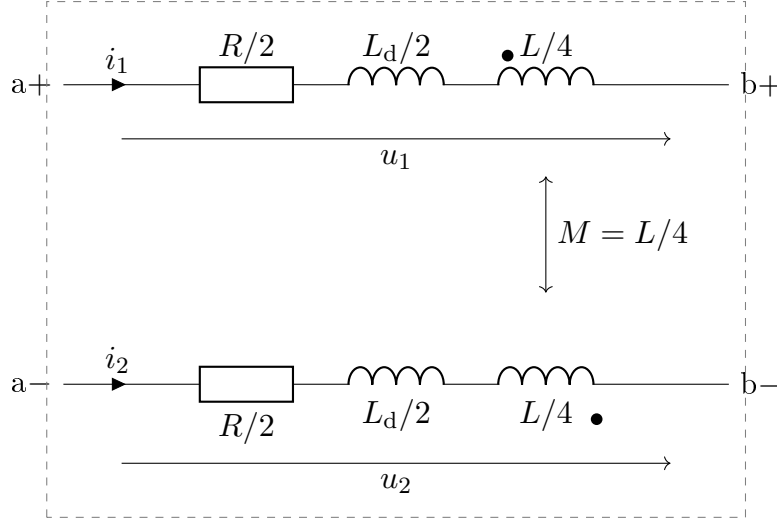


Figure 2: Equivalent circuit of the block in the linear region. The dots mark the coupling: with both currents referred from a to b, the mutual term is $-M$; for the load current ($i_1 = -i_2$) the coupled parts add up. Voltage arrows according to the DIN convention (from higher to lower potential).

The differential mode is the single-phase inductor with saturation; the common mode sees only $R/2$ and $L_d/2$ per conductor and is linear. The two modes are decoupled for any operating point: a common-mode current does not change the saturation state, and saturation does not change the common-mode impedance (within the assumptions of Sec. 7).

2.4 Coupled-inductor form

Eqs. (5)–(7) can be written with an inductance matrix,

$$\begin{bmatrix} u_1 - \frac{R}{2}i_1 \\ u_2 - \frac{R}{2}i_2 \end{bmatrix} = \begin{bmatrix} \frac{L_d}{2} + \frac{L k_L}{4} & -\frac{L k_L}{4} \\ -\frac{L k_L}{4} & \frac{L_d}{2} + \frac{L k_L}{4} \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}, \quad (11)$$

i.e. two coupled inductors with (linear region, $k_L = 1$)

$$L_1 = L_2 = \frac{L_d}{2} + \frac{L}{4}, \quad M = \frac{L}{4}, \quad k = \frac{M}{\sqrt{L_1 L_2}} = \frac{L}{L + 2L_d}. \quad (12)$$

The negative sign of the off-diagonal terms is due to the reference directions (both currents from a to b); with the dots of Fig. 2 the coupling is positive for the load current. The series-aiding inductance $L_1 + L_2 + 2M = L + L_d$ is the inductance seen by the load current, the series-opposing inductance $L_1 + L_2 - 2M = L_d$ is the inductance seen by a common-mode current flowing through both conductors in series. With the default parameters ($L = 300 \mu\text{H}$, $L_d = 20 \mu\text{H}$): $L_1 = L_2 = 85 \mu\text{H}$, $M = 75 \mu\text{H}$, $k = 0.882$. Saturation scales the coupled part only: $L/4$ and M become $k_L L/4$, the leakage $L_d/2$ is unchanged.

The determinant of the matrix in (11) is $\frac{L_d}{2} \left(\frac{L_d}{2} + \frac{L k_L}{2} \right)$, positive for $L_d > 0$: the two currents are then independent states. With $L_d = 0$ the common-mode current would have no inductance and would be fixed algebraically by the external network; the block therefore requires $L_d > 0$.

3 Saturation characteristic

3.1 Differential inductance table

The saturation is described by the differential (incremental) inductance of the main path, normalised to L ,

$$k_L(i_m) = \frac{1}{L} \frac{d\psi}{di_m}, \quad x = \frac{|i_m|}{\sqrt{2} I_{\text{base}}}, \quad (13)$$

as a function of the magnetising current normalised to the peak value of the base current I_{base} (rms). The table is the same as in the single-phase block (Table 1); it is stored as private parameters of the component and is interpolated linearly, with nearest-value extrapolation ($k_L = 0.39$ beyond $x = 3.85$). The characteristic is applied to the magnetising current, i.e. to the net magnetomotive force of the core, not to the individual winding currents.

Table 1: Saturation table (private parameters `Normalized_current`, `L_saturation`) and normalised flux linkage of the main path $\Psi_n(x) = \int_0^x k_L d\xi$ at the breakpoints.

x	k_L	Ψ_n	Ψ_n/x
0	1.00	0	1.000
1.00	1.00	1.000	1.000
1.35	1.00	1.350	1.000
2.30	0.88	2.243	0.975
3.85	0.39	3.227	0.838

3.2 Flux linkage

The flux linkage of the main path follows by integration of the table,

$$\psi(i_m) = \text{sgn}(i_m) L \sqrt{2} I_{\text{base}} \Psi_n(x), \quad \Psi_n(x) = \int_0^x k_L(\xi) d\xi, \quad (14)$$

which is piecewise quadratic and continuous with its first derivative (Fig. 3). The total flux linkage seen by the load current, including the leakage, is

$$\psi_{\text{tot}} = L_d i_m + \psi(i_m), \quad (15)$$

and is the quantity available at the `psi` output of the block (the same as the `psi` output of the single-phase block).

Note that the table describes the *differential* inductance. The apparent (secant) inductance $\psi/i_m = L \Psi_n(x)/x$ decreases much less (Table 1: 84% at $x = 3.85$, where the differential inductance is at 39%). When the table is filled from manufacturer data, check which of the two the data sheet gives: small-signal measurements at DC bias give the differential inductance, $V/(\omega I)$ measurements at rated AC current give a value close to the apparent one.

Setting $I_{\text{base}} = \infty$ gives $x = 0$ and $k_L = 1$ for any current, which disables the saturation.

4 Relation with the single-phase block

For $i_1 = -i_2 = i$, eqs. (5)–(7) give

$$u_1 - u_2 = R i + [L_d + L k_L(i)] \frac{di}{dt}, \quad \psi_{\text{tot}} = L_d i + \int L k_L(i) di, \quad (16)$$

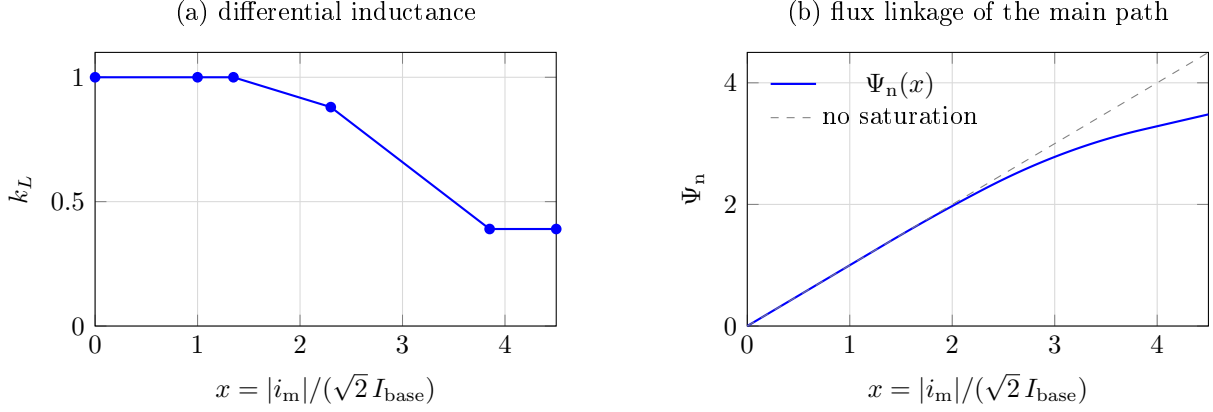


Figure 3: Saturation characteristic with the default table: (a) normalised differential inductance k_L (linear interpolation, nearest-value extrapolation beyond $x = 3.85$); (b) normalised flux linkage of the main path Ψ_n , obtained by integration of (a).

which are exactly the equations of `singlephase_inductor_with_saturation` with the same L , L_d , R and I_{base} . The two-limb block can therefore replace the single-phase block in a two-wire circuit without changing the parameters: the load current sees the same impedance and the same saturation, the voltage drop is split evenly between the two conductors, and in addition the common-mode path is represented by eq. (10).

5 Simscape implementation

5.1 Interface

Table 2 lists the ports, the parameters and the outputs of the component. The port labels a+ and a− are on the left of the block, b+ and b− on the right; the physical-signal outputs are on the right.

Table 2: Interface of `twolimb_inductor_with_saturation`.

Item	Name	Default	Description
node	ap (a+)		winding 1, a side
node	bp (b+)		winding 1, b side
node	an (a−)		winding 2, a side
node	bn (b−)		winding 2, b side
parameter	L	300 μH	main inductance L , total (linear region)
parameter	Ld	20 μH	leakage inductance L_d , total
parameter	R	2.4 $\text{m}\Omega$	winding resistance R , total
parameter	i_base	360 A (rms)	base current I_{base} of the saturation table
output	psi	V s	total flux linkage ψ_{tot} , eq. (15)
output	im	A	magnetising current i_m , eq. (1)

5.2 Variables, equations and states

The component declares the branch currents i_1 (**ap** $\rightarrow$ **bp**) and i_2 (**an** $\rightarrow$ **bn**), the branch voltages u_1 , u_2 , the algebraic variables i_m , x (**i_norm**) and k_L (**kL**), and the flux linkage ψ of the main

path. The equations are (1), the table lookup, (7), (5) and (6), plus the two output equations: ten equations for eight variables and two outputs.

Three variables appear differentiated, i_1 , i_2 and ψ . The two currents are the physical states, with the mass matrix of eq. (11). ψ is a pure integrator of $L k_L di_m/dt$, used only for the **psi** output; its initial value is 0 with high priority, which is consistent with zero initial current. If the network imposes a non-zero initial current, ψ starts with an offset equal to $\psi(i_m(0))$; this affects the output only, not the dynamics. k_L is algebraic and carries no initial-value priority (a high-priority target on an algebraic variable produces an initialisation warning whenever the initial current is above $\sqrt{2} I_{\text{base}}$).

The magnetising current enters the table through $|i_m|$, which introduces a zero crossing at $i_m = 0$. It has no numerical consequence because k_L is flat ($k_L = 1$) up to $x = 1$; a symmetric table without the absolute value would avoid the zero crossing at the cost of duplicating the table.

Parameter checks (**assert**) require $L > 0$, $L_d > 0$, $R \geq 0$ and $I_{\text{base}} > 0$.

5.3 Use

The component is a plain **.ssc** file; it is used either inside a package folder with **ssc_build**, as the other components of the library, or through a *Simscape Component* block pointing at the file. The saturation table is stored as private parameters, so a reactor with a different characteristic needs a copy of the file with the two vectors edited; the breakpoints must be increasing and start at $x = 0$.

The same file describes a common-mode choke by reversing the sense of coil 2: replace $i_m = (i_1 - i_2)/2$ with $(i_1 + i_2)/2$ and the sign of $\frac{1}{2} d\psi/dt$ in eq. (6). The common-mode current then magnetises the core and the load current sees the leakage only.

6 Verification of the equations

The equations (1)–(7) were integrated numerically (variable-step solver, relative tolerance 10^{-9}), independently of Simscape, together with the single-phase equation (9), in the following test network: the terminals a+ and a− are driven by ideal sources referred to ground, $v_{a+} = v_c + v_d/2$ and $v_{a-} = v_c - v_d/2$; the terminals b+ and b− are loaded by $R_\ell = 0.1 \Omega$ each to ground. v_d is a 50 Hz sinusoid of 400 V peak, which drives the load current to 3.7 p.u. (k_L down to 0.45); v_c is a 1 kHz sinusoid of 20 V peak, which drives a common-mode current of 168 A peak. Parameters at their default values.

Fig. 4 shows the steady-state waveforms. The results confirm the properties of Sec. 2:

- the magnetising current i_m of the two-limb model coincides with the current of the single-phase model in the same differential-mode loop within 1×10^{-4} A (peak 1864 A), and the **psi** outputs coincide within 1×10^{-7} V s;
- the common-mode current i_c follows eq. (10) with the external R_ℓ within 2×10^{-7} A; its amplitude $v_c / \sqrt{(R_\ell + R/2)^2 + (\omega L_d/2)^2} = 168$ A is unaffected by the saturation of the differential mode;
- the main-path flux linkage ψ does not respond to v_c .

The check validates the equations and the mode decoupling, not the Simscape syntax: the **.ssc** file was not compiled in the environment in which this note was written and must be built and tested in MATLAB, for instance with the same test network.

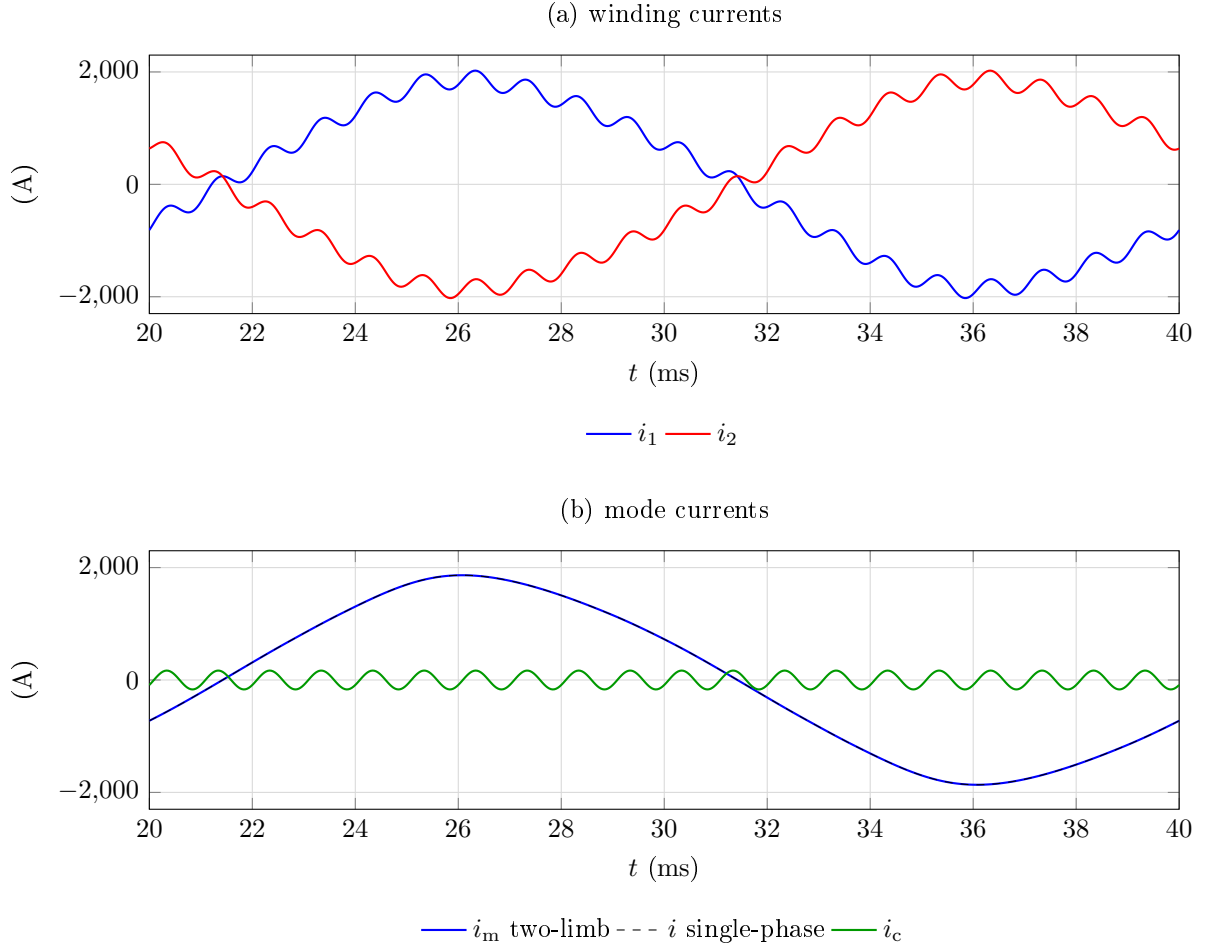


Figure 4: Test with simultaneous differential-mode (400 V, 50 Hz) and common-mode (20 V, 1 kHz) excitation, default parameters: (a) winding currents i_1 , i_2 ; (b) magnetising current i_m compared with the current of the single-phase block in the same loop (curves overlap) and common-mode current i_c . Saturation (x up to 3.7) raises the peak of i_m from 1770 A (linear inductor) to 1864 A.

7 Limitations

- A single saturation factor is computed from the net magnetomotive force: the model assumes the same flux in both limbs (series magnetic circuit). The local flux that each coil drives through its own limb when the currents are not balanced (common-mode or asymmetric currents) is not represented; with large common-mode currents one limb may saturate locally.
- Leakage and common-mode inductance are lumped in the single linear parameter $L_d/2$ per coil. In a real two-limb reactor the common-mode inductance is not necessarily half of the differential-mode leakage.
- No core losses, no winding capacitances, no temperature dependence; R is constant.
- The differential inductance is piecewise linear, so di/dt has kinks at the breakpoints; $\psi(i)$ is continuous with its first derivative.
- Beyond the last breakpoint k_L is held constant (0.39); in deep saturation the differential inductance of a real reactor tends to the air-core value.

References

- [1] R. W. Erickson and D. Maksimović, *Fundamentals of Power Electronics*, 3rd ed. Cham, Switzerland: Springer, 2020, ch. 13–14 (magnetics, coupled inductors).
- [2] M. K. Kazimierczuk, *High-Frequency Magnetic Components*, 2nd ed. Chichester, U.K.: Wiley, 2014.
- [3] The MathWorks, Inc., *Simscape Language Guide*, Natick, MA, USA. [Online]. Available: <https://www.mathworks.com/help/simscape/>

A Source listing

File `twolimb_inductor_with_saturation.ssc` (MIT licence header omitted).

```

1 % Revision history
2 %   Rev. 00 2026-09-18 first issue
3
4 component twolimb_inductor_with_saturation
5 % Two-limb inductor with saturation
6 %
7 % Single-phase reactor with one winding on each limb of a common core:
8 % winding 1 between a+ and b+, winding 2 between a- and b-. The windings
9 % are connected so that the load (differential-mode) current, entering at
10 % a+ and returning at a-, magnetises the core in the same direction. A
11 % common-mode current (same direction in both conductors) does not
12 % magnetise the core and only sees the leakage inductance.
13 %
14 % L, Ld and R are the TOTAL values seen by the load current
15 % (a+ -> b+ -> load -> b- -> a-); half of each is placed in each conductor.
16 % With i1 = -i2 the block is identical to singlephase_inductor_with_saturation
17 % with the same parameters.
18 %
19 % Equivalent coupled-inductor description (linear region):
20 %   L1 = L2 = Ld/2 + L/4,   M = L/4,   k = L/(L + 2*Ld)
21 % The saturation factor kL(im) scales the coupled part (L/4 and M) only,
22 % im = (i1 - i2)/2 being the magnetising current (= load current in
23 % differential mode). The saturation table is the DIFFERENTIAL inductance

```

```

24 %  $d\psi/di$  normalised to  $L$ , versus  $|im|$  normalised to the peak base current
25 %  $\sqrt{2} \cdot i_{base}$ .
26 %
27 % To disable the saturation effect set  $i_{base}$  to  $inf$ .
28
29 nodes
30     ap = foundation.electrical.electrical;    % a+:left
31     an = foundation.electrical.electrical;    % a-:left
32     bp = foundation.electrical.electrical;    % b+:right
33     bn = foundation.electrical.electrical;    % b-:right
34 end
35
36 outputs
37     psi_out = {0, 'V*s'};    %  $\psi$ 
38     im_out  = {0, 'A'};    %  $im$ 
39 end
40
41 parameters
42     L      = {300e-6, 'H'};    % Main (magnetising) inductance, total
43     Ld     = {20e-6, 'H'};    % Leakage inductance, total
44     R      = {2.4e-3, 'Ohm'}; % Winding resistance, total
45     i_base = {360, 'A'};    % Base current (Arms)
46 end
47
48 parameters(Access=private)
49     %  $d\psi/di$  normalised to  $L$ , versus  $|im| / (\sqrt{2} \cdot i_{base})$ 
50     L_saturation = {[1.0, 1.0, 1.0, 0.88, 0.39], '1'};
51     Normalized_current = {[0, 1.0, 1.35, 2.30, 3.85], '1'};
52 end
53
54 variables
55     u1 = {0, 'V'};    %  $u_1$  (a+ to b+)
56     u2 = {0, 'V'};    %  $u_2$  (a- to b-)
57     i1 = {0, 'A'};    %  $i_1$  (a+ -> b+)
58     i2 = {0, 'A'};    %  $i_2$  (a- -> b-)
59     im = {0, 'A'};    % magnetising current  $(i_1 - i_2)/2$ 
60     i_norm = {0, '1'}; %  $|im|$  in p.u. of the peak base current
61     kL = {1, '1'};    % saturation factor
62     % core flux linkage of the main path, total for the loop
63     psi = {value = {0, 'V*s'}, priority = priority.high};
64 end
65
66 branches
67     i1 : ap.i -> bp.i;
68     i2 : an.i -> bn.i;
69 end
70
71 equations
72     assert(L > 0, 'Main inductance must be positive');
73     assert(Ld > 0, 'Leakage inductance must be positive');
74     assert(R >= 0, 'Winding resistance must be non-negative');
75     assert(i_base > 0, 'Base current must be positive');
76
77     u1 == ap.v - bp.v;
78     u2 == an.v - bn.v;
79
80     im == (i1 - i2)/2;
81     i_norm == abs(im)/(i_base*sqrt(2));
82     kL == tablelookup(Normalized_current, L_saturation, i_norm, ...
83         interpolation = linear, extrapolation = nearest);
84
85     % core (coupled) flux linkage, total for the loop:  $d\psi = L \cdot kL(im) \cdot dim$ 
86     psi.der == L*kL*(i1.der - i2.der)/2;

```

```
87
88     % half of the leakage, resistance and core EMF in each conductor
89     u1 == (R/2)*i1 + (Ld/2)*i1.der + psi.der/2;
90     u2 == (R/2)*i2 + (Ld/2)*i2.der - psi.der/2;
91
92     % total flux linkage seen by the load current (same as the single-phase block)
93     psi_out == psi + Ld*im;
94     im_out  == im;
95 end
96 end
```